



Morphology IV

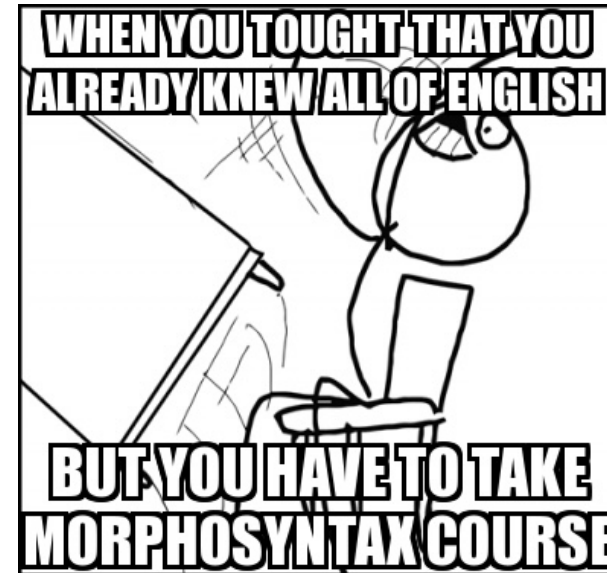
Merlin Balkhash
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2026/09/21 (updated: 2026-09-20)

Please pick up your **name card** from the front

Selectional Restrictions and Order

the **selectional restrictions** of affixes often help us **determine** the order in which they were attached

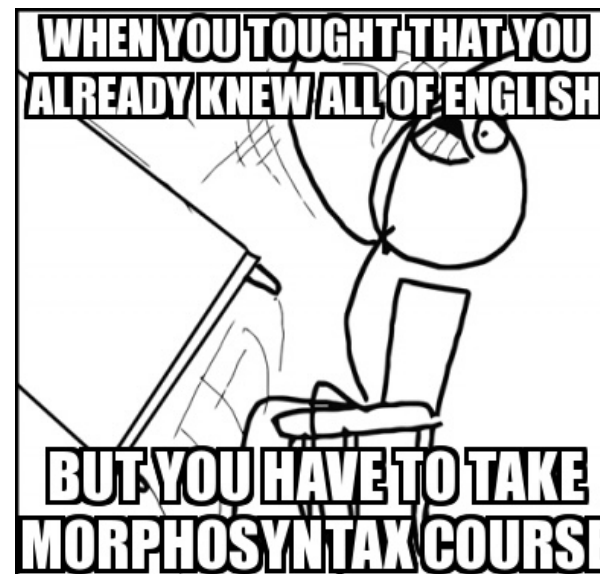
- consider the following polymorphemic word
 - **re-play-able**
- which affix was attached **first** to the root *play*?



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- consider the following polymorphemic word
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- **re-** has to be attached first, since it cannot attach to adjectives
 - *play* → **re-play** → **re-play-able**

Morphological Tree Drawing

De-humid-ify-er

1. identify the word's morphemes and write them out separately in one row
2. identify the root (or roots) and label its **part of speech**
3. look at the morphemes immediately before and after the root
check **which one is added first**, draw lines upward connecting them,
and **label** the new words's part of speech
4. this new word becomes the base you'll keep building off of
repeat step 3 using this new base

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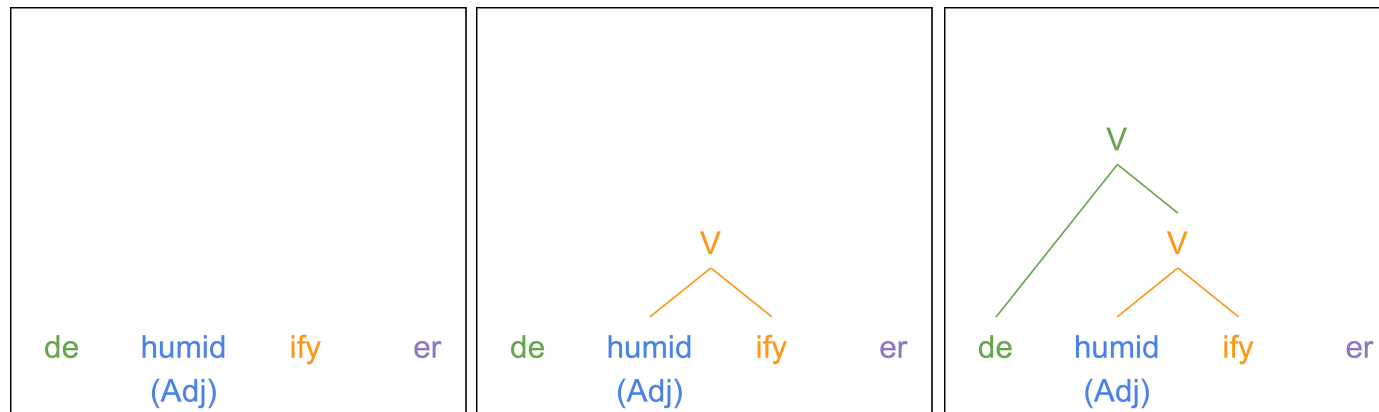
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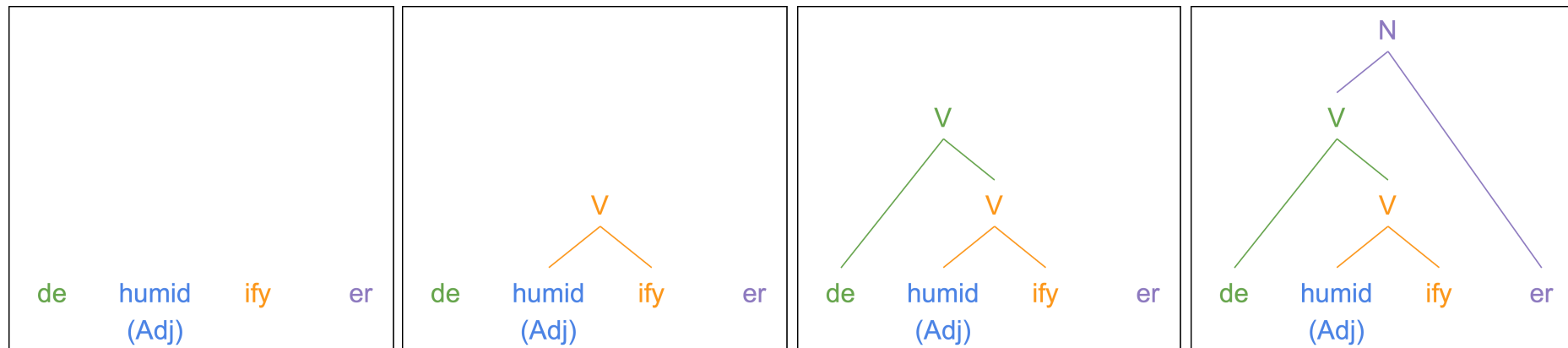
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Morphological Structure: Ambiguity

un + able



undo + able

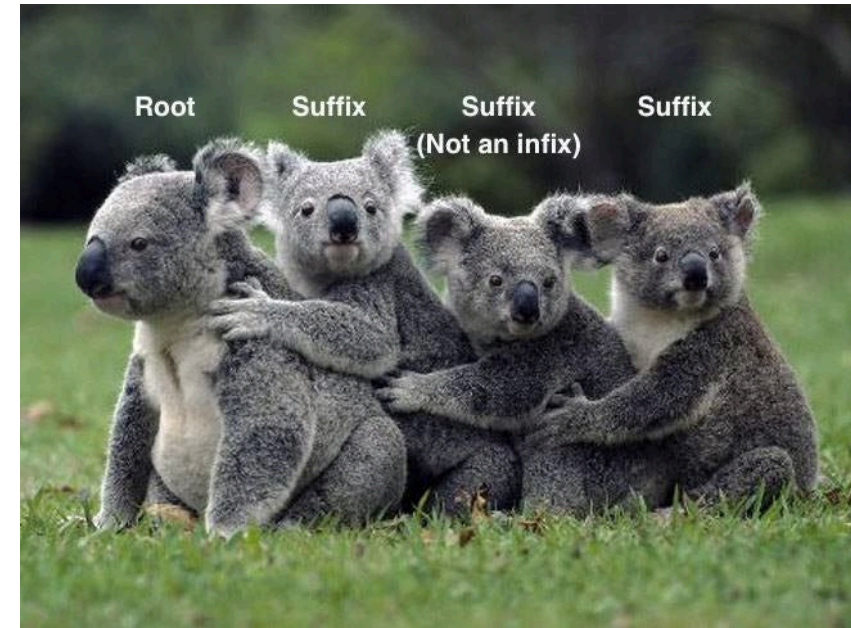


- the case of **unlockable** is an example of a word with an **ambiguous structure**
- depending on the structure you assign it, it has a **different meaning**
- both meanings are acceptable
- which means the **meaning** of words is directly affected by their morphological structure
- that's why **structure** is important, without knowing the structure, we would not be able to be **compositional** properly

Morphological Processes

Morphological Process I: Affixation

- **affixation**: adding an affix (a prefix, suffix, infix, or circumfix)
 - depending on the type of affix, also called **prefixation**, **suffixation**, **infixation**, **circumfixation**
 - **prefixation**
re- + *visit* → *revisit*
 - **suffixation**
modern + *-ize* → *modernize*
 - **infixation**
(Tagalog) *bili* 'buy' → *b-um-ili* 'to buy'
 - **circumfixation**
(Indonesian) *ajar* 'teach' → *pel-ajar-an* 'subject'



Morphological Process II: Compounding

- **compounding**: combining **two or more roots** into a single word
 - *tree + house* → *treehouse*
 - *blue + bird* → *bluebird*
 - *snow + white* → *Snowwhite*
- compound words usually contains more specific meaning than two separate words
- compound words often sound different from the words used separately – one of the roots will be pronounced with **less stress** than if it had been a separate word:
 - *Abraham Lincoln was born in a log cabin – a **tree house**.*
 - *The child played in a **treehouse**.*



Morphological Process II: Compounding

- **compounding**: combining **two or more roots** into a single word
- different languages have different rules for compounding
- **Germanic languages** are well-known for permitting lots of compounding

German: *Donaudampfschiffahrtsgesellschaftskapitän*

Roots: *Donau- dampf- schiff- fahrts- gesellschafts- kapitän*

Meaning: Danube- steam- ship- travel- company- captain
'captain of the Danube Steamship Company'

- **English** is also a Germanic language and allows this kind of compounding
- the meaning of compounds is usually **non-**

compositional



broken vs. broken-broken

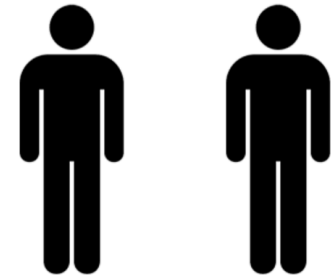
listen to this anecdote about a train breaking down

1. what does the speaker mean when he says the train was **broken vs. broken-broken**?
2. what's the **difference**?

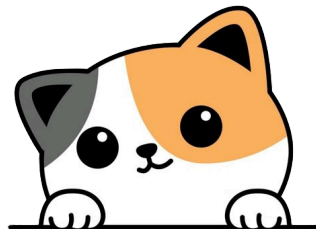
[video](#)



This is me.



Now there are two of me.
This is a meme.

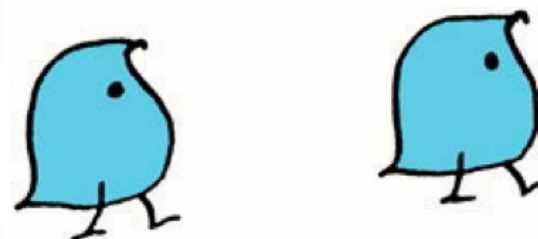


Morphological Process III: Reduplication

- **reduplication**: repeating all or part of a morpheme
- **English** examples:
 - *Do you just like him as a friend, or do you **like-like** him?*
 - *Yesterday, we just went out for coffee, but this weekend we're going on a **date-date**.*
 - *Whenever we go to a **fancy-shmancy** restaurant, we feel like James Bond.*
 - *He's just a baby! **Baby-shmaby**, he's five years old!*



This is a Wug.



Now there is another one.
There are two of them.
There are two Wug Wug.

Morphological Process III: Reduplication



Merriam-Webster 
@MerriamWebster

‘Rhyming reduplication’ is an itsy-bitsy super-duper language quirk that is sure to razzle-dazzle.

Rhyming reduplication appears to make words...sillier?

-hanky-panky
-okey-dokey
-hurley-burley
-hootchy-kootchy
-boogie-woogie

- **reduplication** is more productive in some other languages, serves an explicit **grammatical** purpose:

full reduplication: repeat **ALL** of a morpheme

- **Indonesian** plural
 - *orang* 'person' → *orang-orang* 'people'
- **Chinese** intensified adjective:
 - *kʰaɪ⁵⁵ɕin⁵⁵* 'happy' → *kʰaɪ⁵⁵kʰaɪ⁵⁵ɕin⁵⁵ɕin⁵⁵* 'very happy'

partial reduplication: repeat **PART** of a morpheme

- **Tagalog** future
 - *bili* 'buy' → *bi-bili* 'will buy'
- **Kazakh** intensified adjective:
 - *tætti* 'sweet' → *tæp-tætti* 'very sweet'

Morphological Process IV: Alternation

- **alternation**: changing one or more sounds within the morpheme
- usually most of the sounds stay the same, but one or a few are changed
- alternation can result in *irregular* nouns, verbs, etc.
- what is the difference in **sound** in these words and what is the difference in **meaning**? (inflection or derivation?)
 - goose, **geese**
 - **ring**, **rang**, **rung**
 - Does the baby have **teeth** yet? No, she *hasn't yet started to **teethe***.
 - this tool has no **use**. What are you talking about – I **use** it every day.



Morphological Process V: Suppletion



- **suppletion**: when an inflected form is **phonetically very different** from other forms of the same word (the result is even more highly irregular)
- **English** examples:

PAST TENSE

- **regular**: *walk* → *walked*, *jump* → *jumped*
- **suppletive**: *is* → *was*, *go* → *went*

COMPARATIVE / SUPERLATIVE

- **regular**: *nice* → *nicer* → *nicest*
big → *bigger* → *biggest*
- **suppletive**: *good* → *better* → *best*
bad → *worse* → *worst*

Multiple Processes

some words exhibit **multiple morphological processes**

break → *broke* (alternation)

break → *brok(e)-en* (alternation + suffixing)

Morphological Process VI: Zero Derivation

Calvin and Hobbes

by Bill Watterson



how is Calvin (the child) using language in a creative or unusual way?

is Hobbes (the tiger) correct that Calvin's phrasing is a "complete impediment to understanding"?

Morphological Process VI: Zero Derivation

"I'll change him."



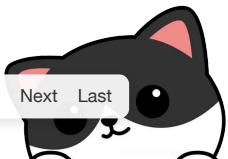
- we know that **derivational** morphemes can change a word's part of speech:
 - *magnet* (N) → *magnet-ize* (V)
 - *atom* (N) → *atom-ic* (Adj)
- English can sometimes change a word's **part of speech** without adding anything
 - *I'm going to **jump*** (V) *over that railing.*
 - *That was a nice **jump*** (N)!
 - *Please send me an **email*** (N) *about that.*
 - *OK, I **emailed*** (V) *you!*
- **zero-derivation**: deriving a new word with a **different part of speech** by adding nothing
- not all languages can do this! how about yours?

Exercise: Morphological Processes

identify the **morphological processes** involved in each of the following words.

affixation (prefixation, suffixation), **compounding**, **reduplication**, **alternation**, **suppletion**

- feed → fed
- leaf → leaves
- peace → peaceful
- foot, ball → football
- good → best
- chair, woman → chairwomen
- back, pack → backpacks
- date → date-dated
- go → underwent
- ox → oxen



Morphological Types of Languages

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languages can be classified according to the way they **use** or **don't use** morphological processes

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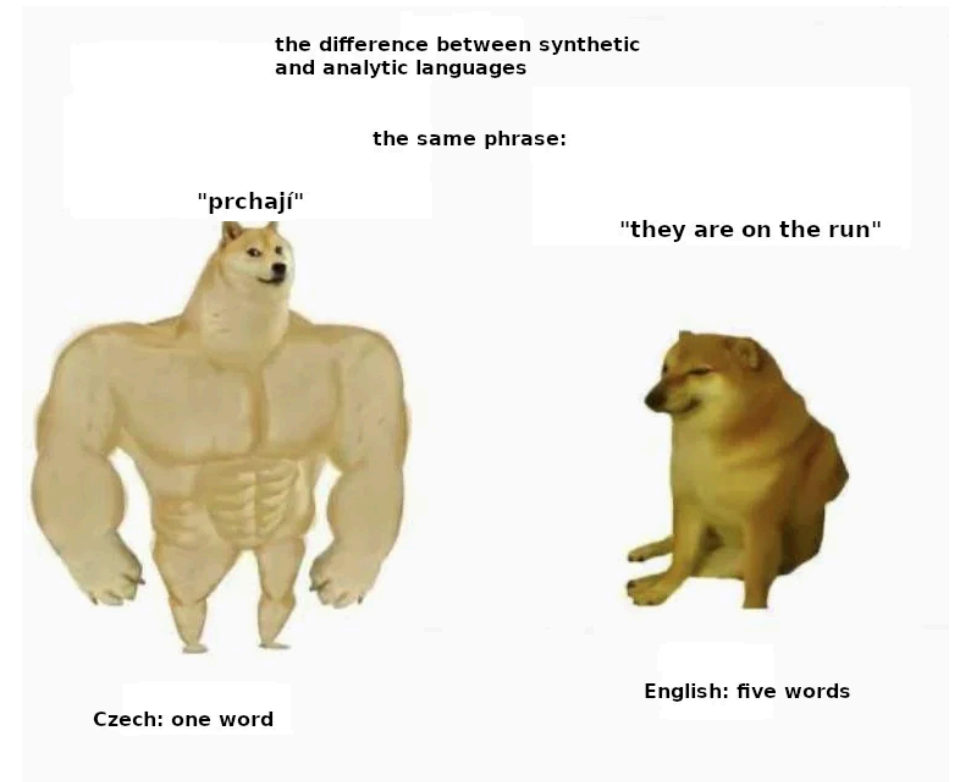
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analytic/isolating language

- made up of sequences of **free morphemes**
- NO **affixes** are used to compose words
- Mandarin Chinese**
 - the past tense is expressed with a functional free morpheme (tones omitted):

(a) [wɔ²¹⁴ mə³⁵n tæ³⁵n tɕi³⁵n]
I plural play piano
 'We are playing the piano.'

(b) [wɔ²¹⁴ mə³⁵n tæ³⁵n tɕi³⁵n] lə
I plural play piano lə
 'We played/have played the piano.'



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synthetic language

- **bound morphemes** can be attached to other morphemes
 - so a word can be made of **several types** of meaningful elements
 - yet there are different ways to put together morphemes, resulting in different types of synthetic languages:
-

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A. **agglutinating**: the morphemes are joined together loosely, it is easy to determine the boundaries between morphemes

- one morpheme has **only one** meaning
- **Japanese**, Turkish, etc.

piano	o	hiki	-ma	-shi	-ta
piano	ACC	play	POL	POL	PST

'We played the piano.'

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B. **fusional**: morphemes are “fused” with the stem and can be hard to tell where the boundary is

- a single affix can convey **several** meanings simultaneously and there can be **alternations** to the forms of both the stems and affixes
- **Greek**, Spanish, etc.

pez	-a	-me		to	piano
play	TH	1PL.PST		the	piano

'We were playing the piano.'

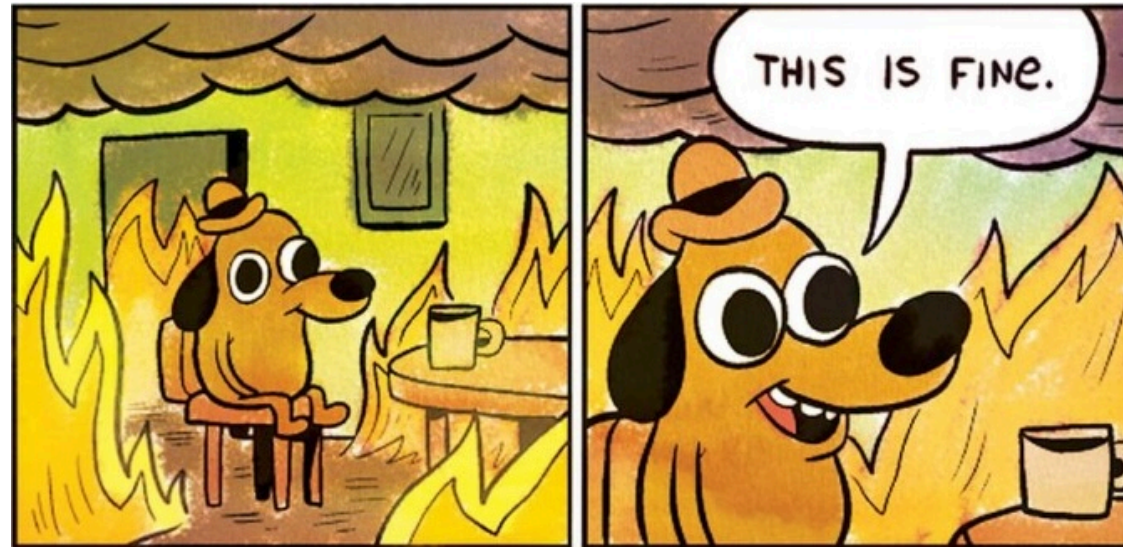
DISCUSSION

Is English **analytic** or **synthetic**, and if it is synthetic, is it **agglutinating** or **fusional**?

Homework I is due this Friday (**Sept 25**)

Homework II will be published before next Wednesday, and it is due next Friday (**Oct 02**)

reading Chapter 2: Part of Speech from *Andrew Carnie's Syntax textbook*



Slides created via the R package **xaringan**.